would be for sewage sludge treatment, which may become increasingly important as the requirements for sterilization and secondary treatment of the sludge increases. The thermoradiation of sludge may also make it useful as a sterile fertilizer.

The low penetrating radiation, long half-life, and high power density of ^{238}Pu makes it ideal for special purpose power supplies (§7.9.3). The main present use is in space research, and $\sim$30 kW_{th} power sources have been launched into space, ^{238}Pu has been used in heart pacemakers and is still a candidate as a power source for completely artificial hearts. Production of ^{238}Pu requires the isolation of ^{237}Np, which is then irradiated and reprocessed to produce pure ^{238}Pu.

The demand for ^{241}Am is larger than present production capacity because of its use in logging oil wells, in smoke detectors, and for various gauging and metering devices. However, the potential source is large. At the end of 2003, the amount of americium produced as a result of civil nuclear power reactor operation was about 87 tonnes. About 68 tonnes of this americium resulted from the decay of ^{241}Pu subsequent to its discharge from the reactors. Currently, the americium inventory increases by about 4 tonnes per year.

If all the mentioned waste products are recovered, this would mean (a) that the waste is turned into an essential asset with benefits in food production, health, and safety, and (b) that the hazard of the remaining waste would be much lower, considerably simplifying final waste storage.

21.15. EXERCISES

21.1. In a BWR the minimum and maximum heat fluxes at the fuel rod surface are 0.46 and 1.08 MW m^{-2} at a coolant temperature of 283°C. The rods have an outer diameter of 12.7 mm with a cladding of 0.8 mm thick Zircaloy-4; assume a negligible fuel cladding gap and neglect the temperature drop between coolant and cladding and across the fuel-cladding gap. Assume that data in Table 21.1 are valid at all temperatures and $k_{UO2} = 4$ W/m°C. What are the highest and lowest fuel-center temperatures?

21.2. Using the data for the Würgassen reactor and the thermal neutron capture cross-section (Table 19.4) it can be calculated how many kg Pu should be formed per t U at a burn-up of 27 500 MWd/t. (a) Make this calculation assuming that plutonium disappears only through fission in ^{239}Pu. (b) According to Table 21.2 each t U from a PWR contains 8.69 kg Pu; why is your result much lower?

21.3. It is desired that 98% of all ^{233}Th formed by neutron capture in ^{232}Th decays to ^{233}U. How long a time must elapse between end of irradiation and start of reprocessing?

21.4. In a ^{233}U fuelled reactor, some ^{233}U is converted into ^{235}U. Calculate the amount of ^{235}U formed in 1 t ^{233}U from neutron capture in ^{233}U and ^{234}U ($\sigma_{n,\gamma}$

97 b) for a fluence of 10^{25} n m^{-2} (a) assuming no consumption of ^{235}U formed, (b) taking ^{235}U fission and capture into account.

21.5. Explain why ^{228}Th and ^{232}U is a nuisance in the thorium-uranium fuel cycle.

21.6. A reactor starting with 3% ^{235}U produces 6000 MWd energy/t U fuel each year. Neglecting fission in ^{238}U, (a) how much fission products have been produced after 5 years? (b) What is the ^{235}U concentration if plutonium fission also is taken into account?

21.7. In the example above, 1 t U as fuel elements is removed from the reactor after 2 years. Using Fig. 21.7, (a) what is the total radioactivity from the fission products after 1 y cooling time? (b) Which FP elements are the most radioactive ones at this time?

21.8. Calculate the decontamination factor required for (a) fission product activity, and (b) for gadolinium in commercial plutonium nitrate produced from PWR fuel (Tables 21.2 and 21.7) at t_{cool} 10 y.

21.9. 0.0015 Ci ^{239}Pu is released annually from a reprocessing plant. What will be the corresponding release of ^{238}Pu and ^{240}Pu for typical isotopic plutonium composition of LWR fuel?

21.10. Calculate the natural radiotoxicity value In_w of 1 km^3 of land (density 2 600 kg m^{-3}) containing 3 weight ppm ^{238}U with daughter products. Only ^{226}Ra has to be considered.

21.11. A tank contains 100 m^3 5 y old HLLW. Analyses show that a 1 ml sample contains 1.09 GBq of ^{90}Sr, which is the only Sr activity. (a) Calculate the heat production for a waste of composition in Table 21.9 left column. (b) How many 500 kg glass cylinders would be needed (assume the glass contains 10% FPs) to contain all the solidified waste? (c) How many 1000 MW_e PWR reactor years does this waste correspond to?

21.12. Assume that Ru, Rh, and Pd are recovered from the waste from a 10 GW_e program. What will the annual amounts and specific radioactivities be at $t_{cool} =$ 10 y for each of them?

21.16. LITERATURE

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